Cryptographic primitives for quantum networks

Iordanis Kerenidis









Cryptography with distrustful parties

Quantum Networks

Secure and efficient network of quantum and classical untrusted agents who communicate, perform distributed tasks, delegate computation to large-scale servers, etc.



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Fundamental primitives

- Coin Flipping
- Bit Commitment
- Oblivious Transfer
- …

Security models

- Information Theoretic
- Computational
- Device Independent
- Relativist



Outline

1. Bit Commitment (BC)
   a) A simple protocol with cheating probability ¾
   b) A lower bound via Fidelity/Trace Distance
2. Strong Coin Flipping (SCF)
   a) Bit-commitment based coin flipping with ¾
   b) A lower bound via Semi-Definite Programming
   c) Implementations
3. Weak Coin Flipping (WCF)
   a) Kitaev's point game formalism
   b) Optimal protocols for BC and SCF
4. More security models
   a) Device Independent: BC via the GHZ game
   b) Noisy storage: Weak String Erasure
   c) Relativistic: BC protocol
5. More primitives

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   a) A simple protocol with cheating probability ¾ [Ambainis 01]
   b) A lower bound via Fidelity/Trace Distance [Chailloux, K. 11]
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1. Bit Commitment

Commit phase

- Alice commits to a bit x.
- Bob should have no/little information about x.

Reveal phase

- If Alice committed honestly to x, she can reveal x.
- Cheating Alice cannot reveal both values for x (a little weaker than classical definition)

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Security

- Cheating probability for Alice: $P_A^* = \frac{1}{2}(\Pr[\text{reveal } 0] + \Pr[\text{reveal } 1]) \geq \frac{1}{2}$
- Cheating probability for Bob: $P_B^* = \Pr[\text{guess } x \text{ before reveal}] \geq \frac{1}{2}$
- Cheating probability of the protocol: $\max\{P_A^*, P_B^*\}$

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Main ideas

- Commit by sending a quantum state.
- Two non-orthogonal states cannot be perfectly distinguished
- A quantum state cannot be perfectly projected onto two different states

1. Bit Commitment: A simple protocol

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- Alice creates the bipartite state $|\psi_x\rangle = \frac{1}{\sqrt{2}}|xx\rangle + \frac{1}{\sqrt{2}}|22\rangle$ and sends the second system to Bob.
- Bob's state: $\rho_x = \begin{cases} |x\rangle \text{ with prob. } ½ \\ |2\rangle \text{ with prob. } ½ \end{cases}$

Reveal phase

- Alice reveals x and sends the first system of $|\psi_x\rangle$
- Bob checks that Alice did not cheat by measuring.

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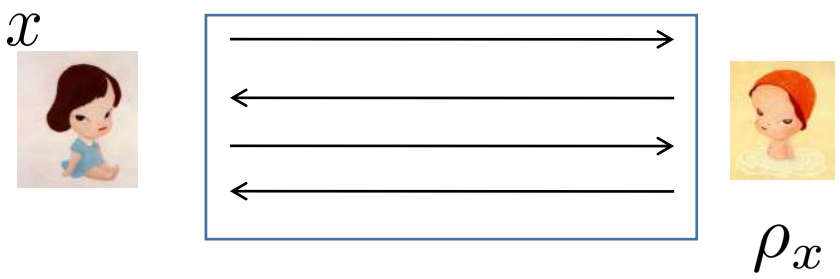
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- Theorem: Alice's optimal strategy is to send the above state.

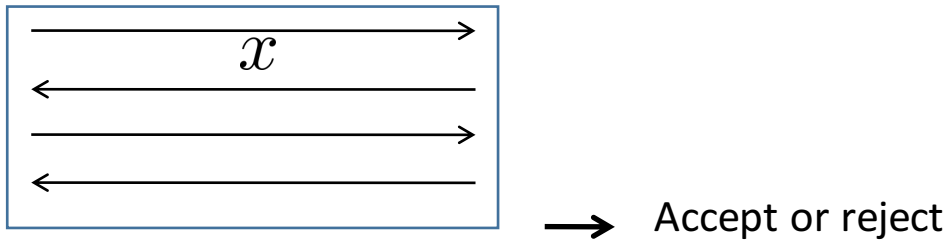
1. Bit Commitment: Lower Bound

Consider any QBC protocol

Commit phase



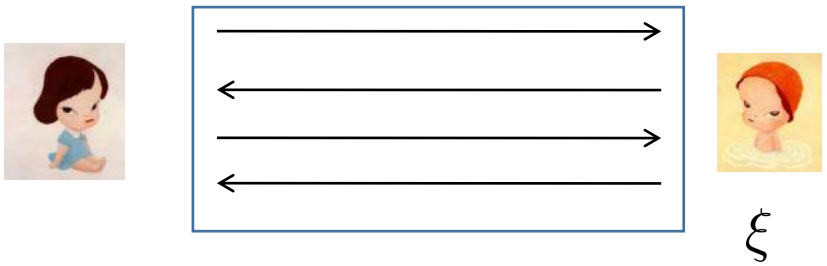
Reveal phase



1. Bit Commitment: Lower Bound (Alice)

Cheating Alice

Commit phase

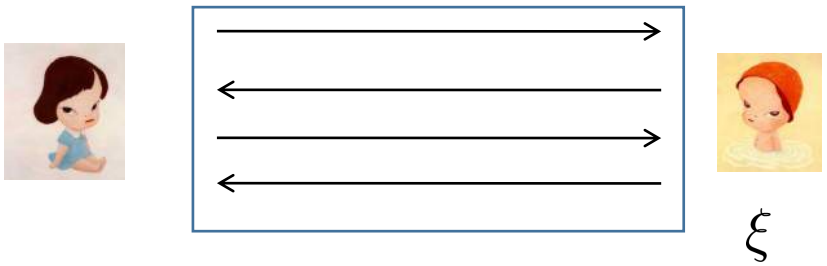


Theorem (Uhlmann) : $\Pr[\text{ Alice successfully reveals } x] \geq F^2(\xi, \rho_x)$

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ξ

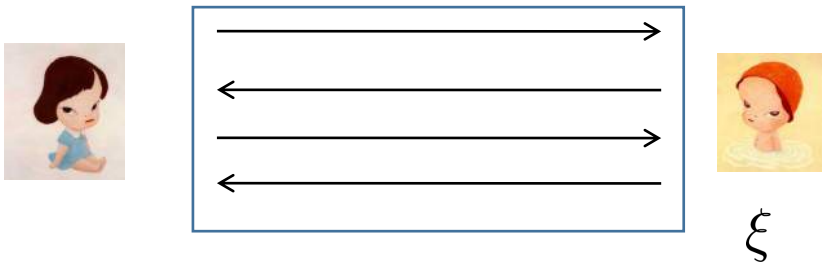
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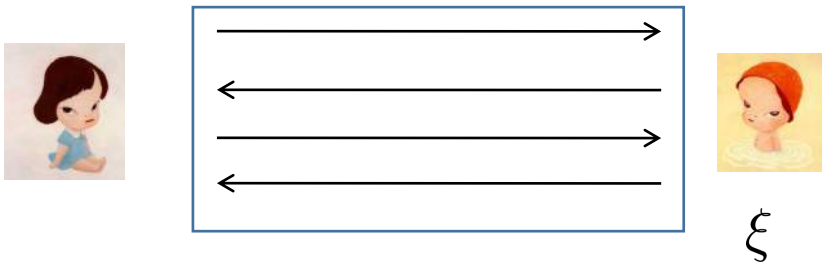
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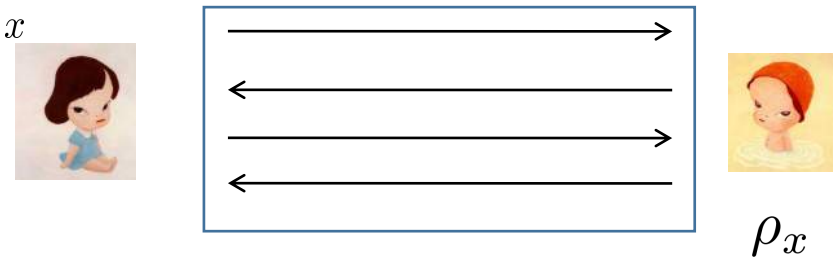
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$$P_A^* = \frac{1}{2}(\Pr[\text{reveal } 0] + \Pr[\text{reveal } 1]) \geq \frac{1}{2}(F^2(\rho_0, \rho_+) + F^2(\rho_1, \rho_+))$$

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Cheating Bob

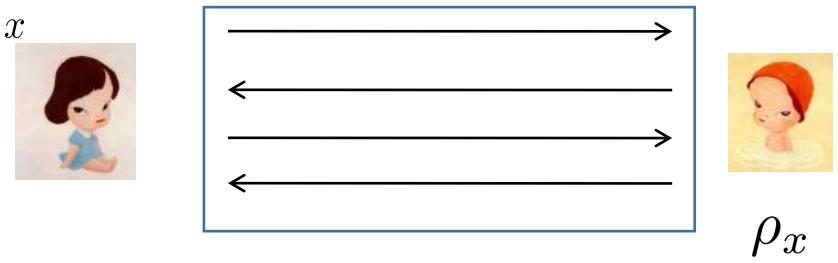
Commit phase



1. Bit Commitment: Lower Bound (Bob)

Cheating Bob

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Theorem (Helstrom): $P_B^* = \Pr[\text{Bob guesses } x] = \frac{1}{2} + \frac{\Delta(\rho_0,\rho_1)}{2}$

1. Bit Commitment: Lower Bound (Recap)

For any QBC protocol

$$P_A^* \geq \frac{1}{2}\left(F^2(\rho_+, \rho_0) + F^2(\rho_+, \rho_1)\right)$$

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Main Lemma: $\forall\ \rho_0, \rho_1 \quad \frac{1}{2}\left(F^2(\rho_+, \rho_0) + F^2(\rho_+, \rho_1)\right) \geq (1 - (1 - \frac{1}{\sqrt{2}})\Delta(\rho_0, \rho_1))^2$

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From there,

$$P_A^* \geq (1 - (1 - \frac{1}{\sqrt{2}})\Delta(\rho_0, \rho_1))^2$$

$$P_B^* \geq \frac{1}{2} + \frac{\Delta(\rho_0, \rho_1)}{2}$$

$$\max\{P_A^*, P_B^*\} \geq \frac{1}{2} + \frac{-5+2\sqrt{2}+\sqrt{21-12\sqrt{2}}}{-12+8\sqrt{2}} \approx 0.739$$

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Proof

$\frac{1}{2}\left(F^2(\rho_+,\rho_0) + F^2(\rho_+,\rho_1)\right)$

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$$\sigma_0 = \frac{1}{2}|0\rangle\langle 0| \otimes \rho_0 + \frac{1}{2}|1\rangle\langle 1| \otimes \rho_1$$

$$\sigma_+ = \frac{1}{2}|0\rangle\langle 0| \otimes \rho_+ + \frac{1}{2}|1\rangle\langle 1| \otimes \rho_+$$

$$\Delta(\sigma_0,\sigma_+) = \Delta(\rho_0,\rho_1)/2$$

$$\frac{1}{2}(F(\rho_+,\rho_0) + F(\rho_+,\rho_1)) = F(\sigma_0,\sigma_+)$$

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Main Lemma: $\forall\ \rho_0,\rho_1 \quad \frac{1}{2}\left(F^2(\rho_+,\rho_0) + F^2(\rho_+,\rho_1)\right) \geq (1-(1-\frac{1}{\sqrt{2}})\Delta(\rho_0,\rho_1))^2$

Proof

$$\frac{1}{2}\left(F^2(\rho_+,\rho_0) + F^2(\rho_+,\rho_1)\right) \geq F^2(\sigma_0,\sigma_+) \geq F^2(D_0,D_+)$$

$$\geq F^2(T_0,T_+) = (1-\left(1-\tfrac{1}{\sqrt{2}}\right)2\Delta(T_0,T_+))^2$$

$$\geq (1-\left(1-\tfrac{1}{\sqrt{2}}\right)2\Delta(D_0,D_+))^2$$

$$\geq (1-\left(1-\tfrac{1}{\sqrt{2}}\right)\Delta(\rho_0,\rho_1))^2$$

4.

$$D_0 = \sum_i p_i |i\rangle\langle i|$$

$$D_+ = \sum_i q_i |i\rangle\langle i|$$

$$F(\sigma_0,\sigma_+) = \sum_i \sqrt{p_i q_i}$$

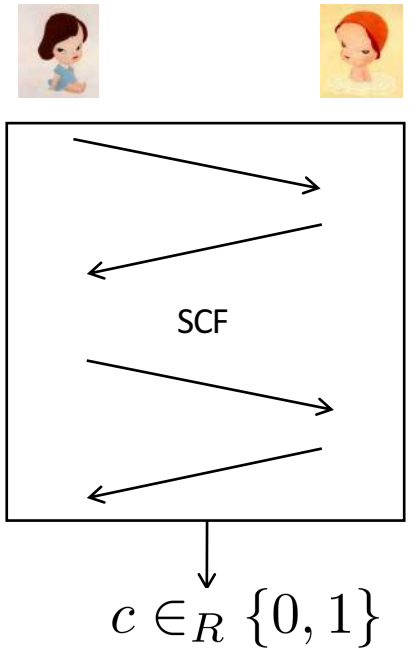
$$\Delta(D_0,D_+) \leq \Delta(\sigma_0,\sigma_+) = \frac{\Delta(\rho_0,\rho_1)}{2}$$

Outline

1. Bit Commitment
   a) A simple protocol with cheating probability ¾
   b) A lower bound via Fidelity/Trace Distance
2. Strong Coin Flipping
   a) Bit-commitment based coin flipping with ¾ [Ambainis 01]
   b) A lower bound via Semi-Definite Programming [Kitaev 03]
   c) Implementations
3. Weak Coin Flipping
   a) Kitaev's point game formalism
   b) Optimal protocols for BC and SCF
4. More security models
   a) Device Independent: BC via the GHZ game
   b) Noisy storage: Weak String Erasure
   c) Relativistic: BC protocol
5. More primitives

2. Strong Coin Flipping

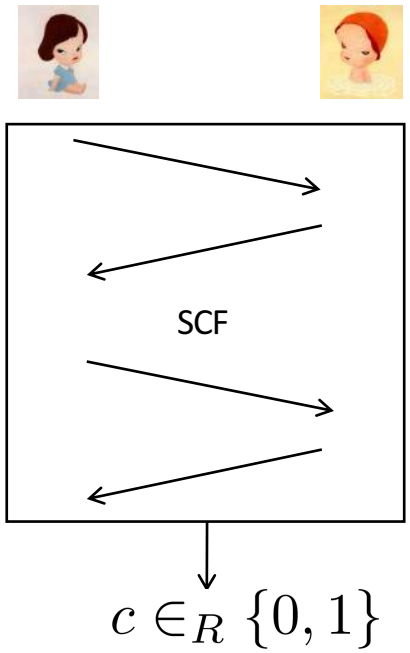
Strong coin flipping with bias ε



- Alice and Bob are far from each other and don't trust each other
- They want to flip a coin c
- They want to make sure that the outcome of c is random

2. Strong Coin Flipping

Strong coin flipping with bias ε



- Alice and Bob are far from each other and don't trust each other
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- They want to make sure that the outcome of c is random

Security

Cheating probability for Alice

$$P_A^* = max_A\{\Pr[c=0], \Pr[c=1]\} \leq \frac{1}{2} + \varepsilon$$

Cheating probability for Bob

$$P_B^* = max_B\{\Pr[c=0], \Pr[c=1]\} \leq \frac{1}{2} + \varepsilon$$

Cheating probability of the protocol

$\max\{P_A^*, P_B^*\}$

2. Strong Coin Flipping: History

– Perfect coin flipping is impossible, i.e. $\varepsilon > 0$ [Lo-Chau, Mayers 96]

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– Better than classical protocols exist (classically always $\varepsilon = 0.5$)

  – $P^* \geq 0.91$ [Aharonov, Ta-Shma, Vazirani, Yao STOC '00]

  – $P^* \geq \frac{3}{4}$ [Ambainis STOC '01]

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- $P^* \geq \frac{3}{4}$ [Ambainis STOC '01]
- $P_A^* P_B^* \geq \frac{1}{2}$ [Kitaev '03] , $P^* \geq \frac{1}{\sqrt{2}}$

2. Strong Coin Flipping: from Bit Commitment

Protocol [Ambainis]

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1. Alice creates the bipartite state $|\psi_x\rangle = \frac{1}{\sqrt{2}}|xx\rangle + \frac{1}{\sqrt{2}}|22\rangle$ and sends the second system to Bob.

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Security

Bob's cheating: $\Pr[\text{guess } x] = \frac{3}{4}$

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Security

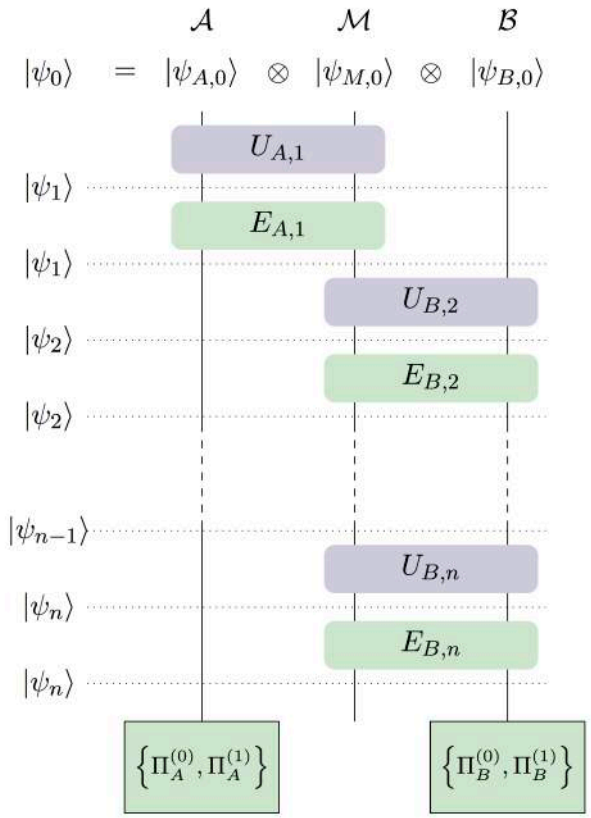
Bob's cheating: $\Pr[\text{guess } x] = \frac{3}{4}$

Alice's cheating: $\frac{1}{2}\Pr[\text{reveal } 0] + \frac{1}{2}\Pr[\text{reveal } 1] = \frac{3}{4}$

2. Strong Coin Flipping: Lower Bound

Protocol [Kitaev]

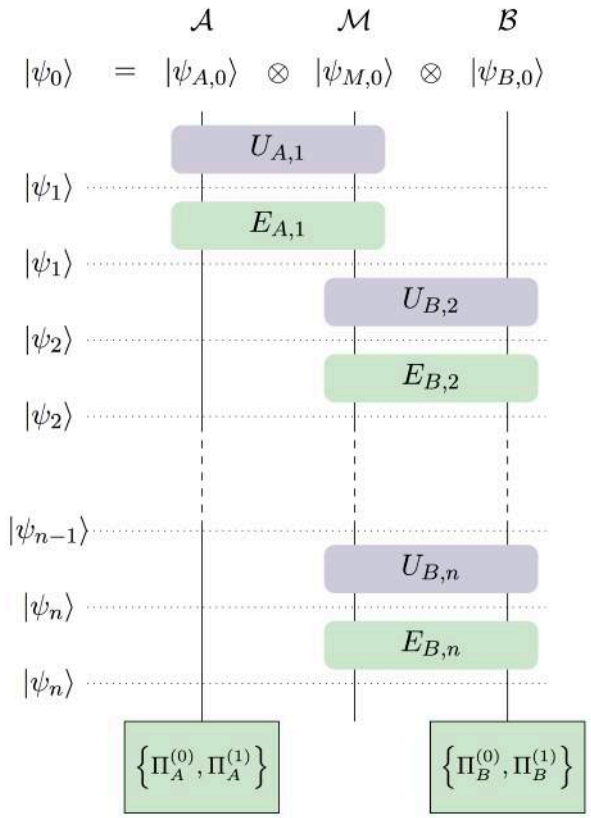
- Three spaces: A, M, B



2. Strong Coin Flipping: Lower Bound

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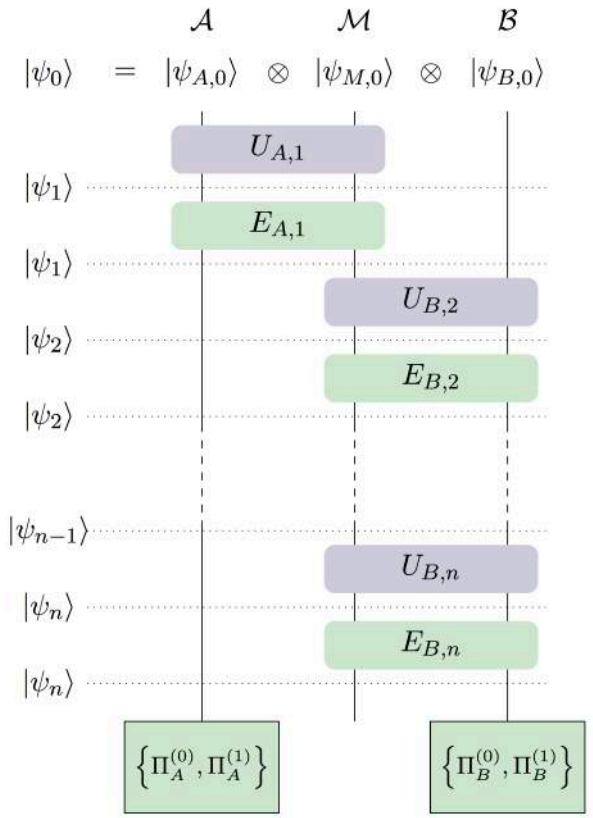
- Three spaces: A, M, B
- Initial state $|\psi_0\rangle$



2. Strong Coin Flipping: Lower Bound

Protocol [Kitaev]

- Three spaces: A, M, B
- Initial state $|\psi_0\rangle$
- Round i
  - Honest states $|\psi_i\rangle$
  - Unitary $U_{A,i}$ / $U_{B,i}$
  - Projection $E_{A,i}$ / $E_{B,i}$



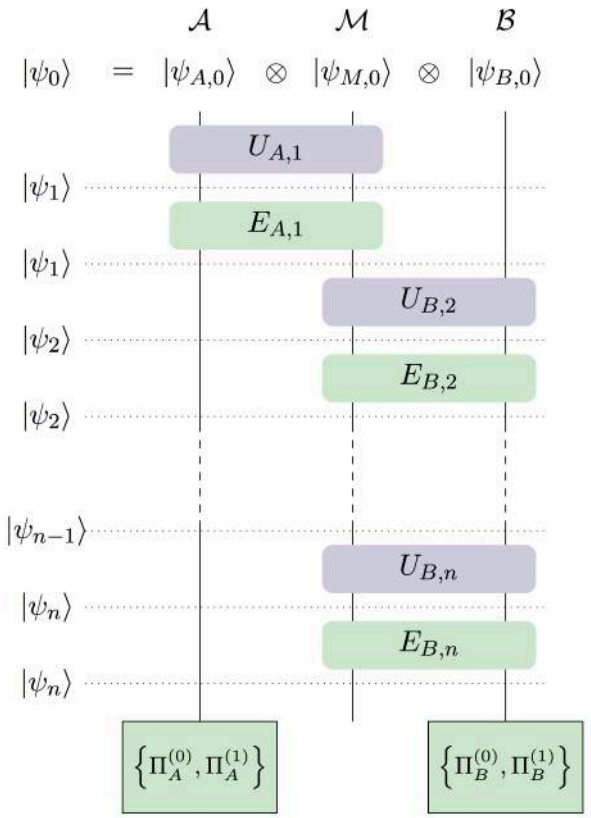
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- Final Projections

$$\langle\psi_n|\Pi_A^{(0)}\otimes I\otimes\Pi_B^{(0)}|\psi_n\rangle = 1/2$$

$$\langle\psi_n|\Pi_A^{(1)}\otimes I\otimes\Pi_B^{(1)}|\psi_n\rangle = 1/2$$

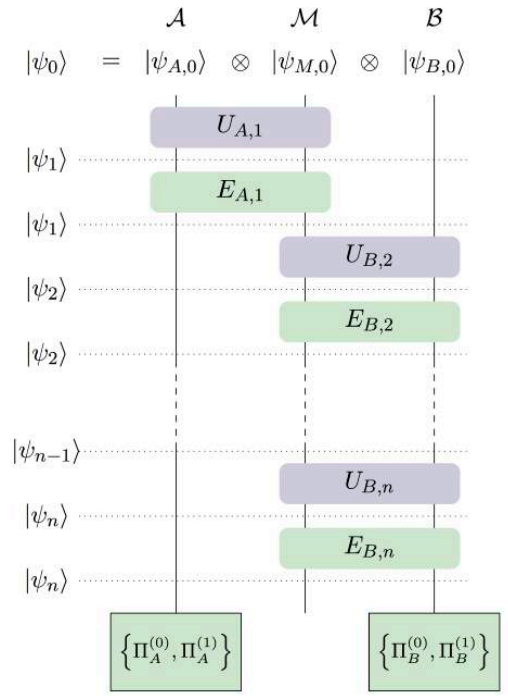


2. Strong Coin Flipping: Lower Bound (SDPs)

Theorem 1 (Primal)

$P_B^* = \max \mathrm{Tr}((\Pi_A^{(1)} \otimes \mathbb{I}_{\mathcal{M}})\rho_{AM,n})$ *over all* $\rho_{AM,i}$ *satisfying*

- $\mathrm{Tr}_{\mathcal{M}}(\rho_{AM,0}) = \mathrm{Tr}_{\mathcal{MB}}(|\psi_0\rangle\langle\psi_0|) = |\psi_{A,0}\rangle\langle\psi_{A,0}|$;
- *for* i *odd,* $\mathrm{Tr}_{\mathcal{M}}(\rho_{AM,i}) = \mathrm{Tr}_{\mathcal{M}}(E_i U_i \rho_{AM,i-1} U_i^\dagger E_i)$;
- *for* i *even,* $\mathrm{Tr}_{\mathcal{M}}(\rho_{AM,i}) = \mathrm{Tr}_{\mathcal{M}}(\rho_{AM,i-1})$.



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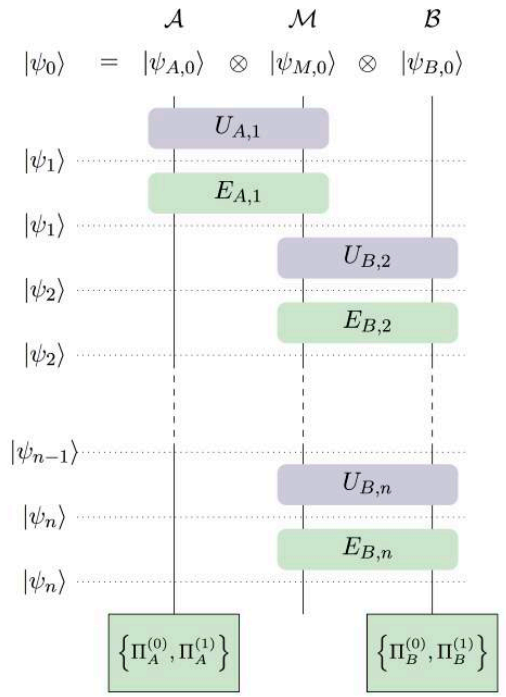
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$P_A^* = \max \mathrm{Tr}((\mathbb{I}_{\mathcal{M}} \otimes \Pi_B^{(0)})\rho_{MB,n})$ *over all* $\rho_{BM,i}$ *satisfying*

- $\mathrm{Tr}_{\mathcal{M}}(\rho_{MB,0}) = \mathrm{Tr}_{\mathcal{AM}}(|\psi_0\rangle\langle\psi_0|) = |\psi_{B,0}\rangle\langle\psi_{B,0}|$;
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2. Strong Coin Flipping: Lower Bound (Dual SDPs)

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Theorem 2 (Dual)

$P_B^* = \min \mathrm{Tr}(Z_{A,0}|\psi_{A,0}\rangle\langle\psi_{A,0}|)$ *over all* $Z_{A,i}$ *under the constraints:*

① $\forall i,\ Z_{A,i} \succeq 0$;

② *for* i *odd,* $Z_{A,i-1} \otimes \mathbb{I}_{\mathcal{M}} \succeq U_{A,i}^\dagger E_{A,i}(Z_{A,i} \otimes \mathbb{I}_{\mathcal{M}})E_{A,i}U_{A,i}$;

③ *for* i *even,* $Z_{A,i-1} = Z_{A,i}$;

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2. Strong Coin Flipping: Lower Bound (Dual Optimal Points)

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$$\left\langle \psi_n \middle| \Pi_A^{(1)} \otimes I \otimes \Pi_B^{(1)} \middle| \psi_n \right\rangle = 1/2$$

2. Strong Coin Flipping: Lower Bound (Dual Optimal Points)

Dual optimal point: $\{Z_{A,i}\},\{Z_{B,i}\}$ s.t.

$$P_A^* = \langle\psi_0|I_A \otimes I_M \otimes Z_{B,0}|\psi_0\rangle$$
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$$P_B^* = \langle\psi_0|Z_{A,0} \otimes I_M \otimes I_B|\psi_0\rangle$$

Then,

$$\begin{aligned} P_A^* \cdot P_B^* &= \langle\psi_0|Z_{A,0} \otimes I_M \otimes Z_{B,0}|\psi_0\rangle \\ &\geq \ldots \\ &\geq \langle\psi_n|Z_{A,n} \otimes I_M \otimes Z_{B,n}|\psi_n\rangle \\ &= \langle\psi_n|\Pi_A^{(1)} \otimes I_M \otimes \Pi_B^{(1)}|\psi_n\rangle \\ &= \frac{1}{2} \end{aligned}$$

Theorem 2 (Dual)

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$$\left\langle \psi_n \middle| \Pi_A^{(1)} \otimes I \otimes \Pi_B^{(1)} \middle| \psi_n \right\rangle = 1/2$$

1.& 2. Recap

	Lower Bound	Upper Bound
Strong Coin Flipping	$\frac{1}{\sqrt{2}}$	3/4
Bit Commitment	0.739…	3/4

What are the optimal protocols?

Outline

1. Bit Commitment (BC)
   a) A simple protocol with cheating probability ¾
   b) A lower bound via Fidelity/Trace Distance
2. Strong Coin Flipping (SCF)
   a) Bit-commitment based coin flipping with ¾
   b) A lower bound via Semi-Definite Programming
   c) Implementations
3. Weak Coin Flipping (WCF)
   a) Kitaev's point game formalism [Mochon 07]
   b) Optimal protocols for BC and SCF [Chailloux, K. 09 and 11]
4. More security models
   a) Device Independent: BC via the GHZ game
   b) Noisy storage: Weak String Erasure
   c) Relativistic: BC protocol
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3. Weak Coin Flipping

– Weak coin flipping with arbitrarily small bias is possible
[Mochon 07, Aharonov, Chailloux, Ganz, K., Magnin 2014]

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– Beautiful techniques for lower bounds and constructions

– Weak Coin Flipping: Master Primitive
  – Optimal weak coin implies optimal strong coin [Chailloux, K. ’09]
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3. Weak Coin Flipping

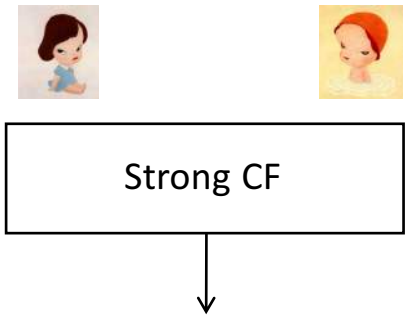
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  – Optimal weak coin implies optimal strong coin [Chailloux, K. ’09]
  – Optimal weak coin implies optimal bit commitment [Chailloux, K. ’11]

– No idea what the protocols look like!

3. Weak Coin Flipping: definition

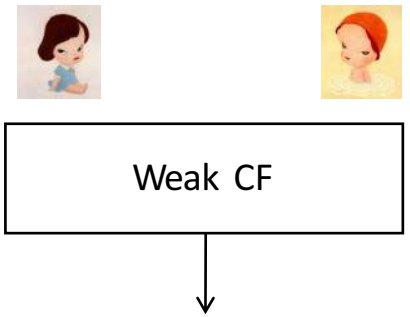


$c \in_R \{0,1\}$

$$P_A^* = max_A\{\Pr[c=0], \Pr[c=1]\} \leq \frac{1}{2} + \varepsilon$$

$$P_B^* = max_A\{\Pr[c=0], \Pr[c=1]\} \leq \frac{1}{2} + \varepsilon$$

$$\max\{P_A^*, P_B^*\}$$



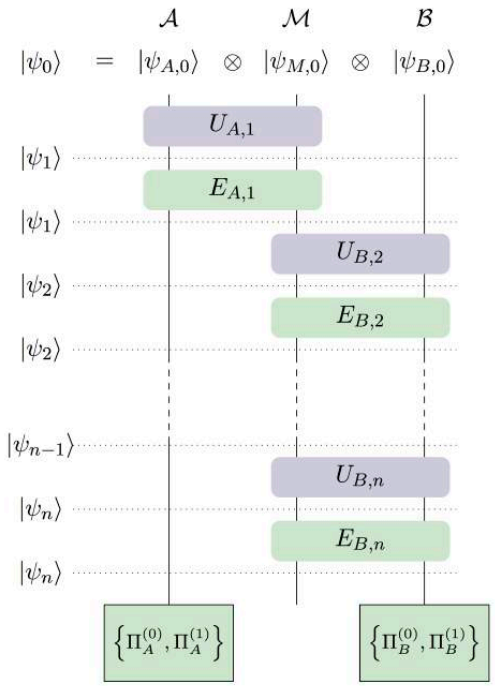
0: Alice wins / 1: Bob wins

$$P_A^* = max_A\{\Pr[c=0]\} \leq \frac{1}{2} + \varepsilon$$

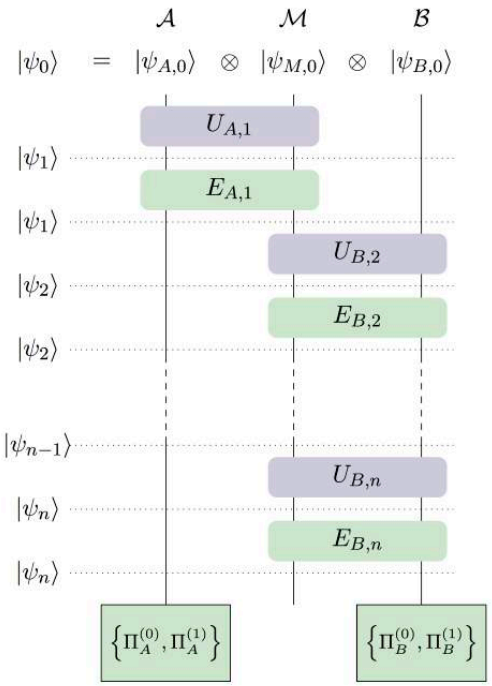
$$P_B^* = max_A\{\Pr[c=1]\} \leq \frac{1}{2} + \varepsilon$$

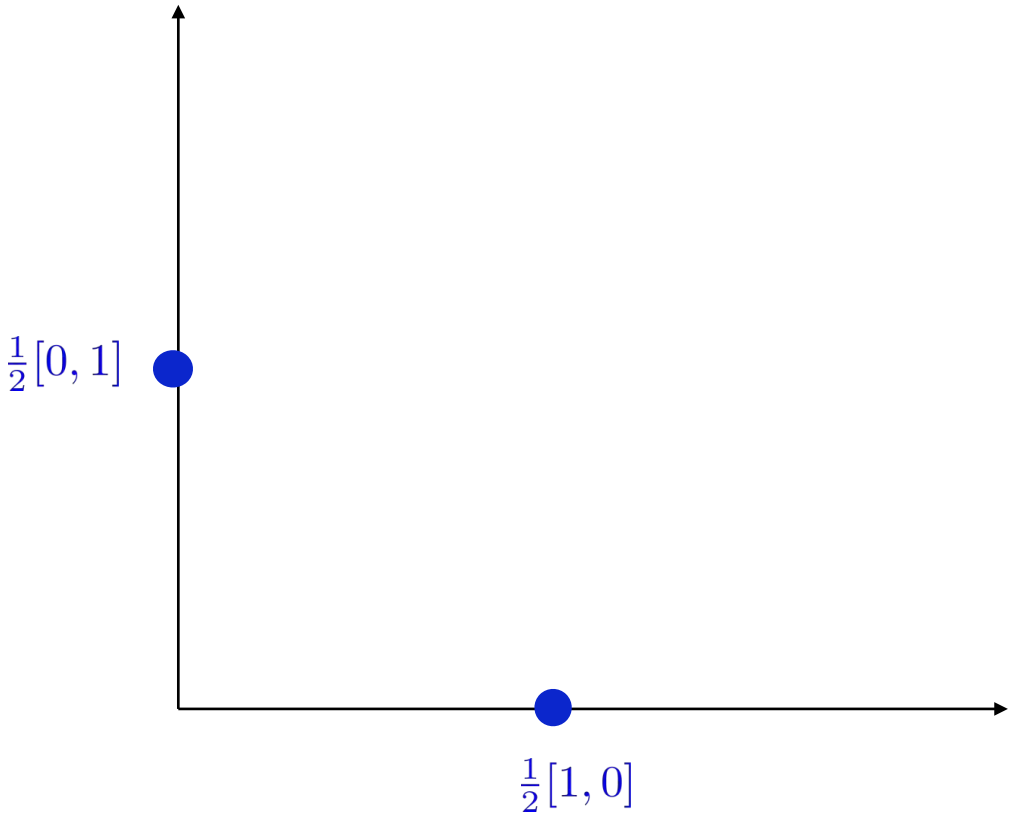
$$\max\{P_A^*, P_B^*\}$$

3. Weak Coin Flipping: From SDP to Point Games

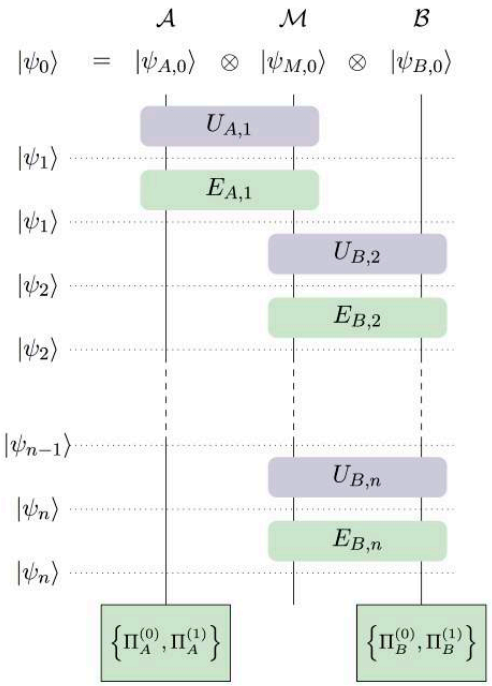


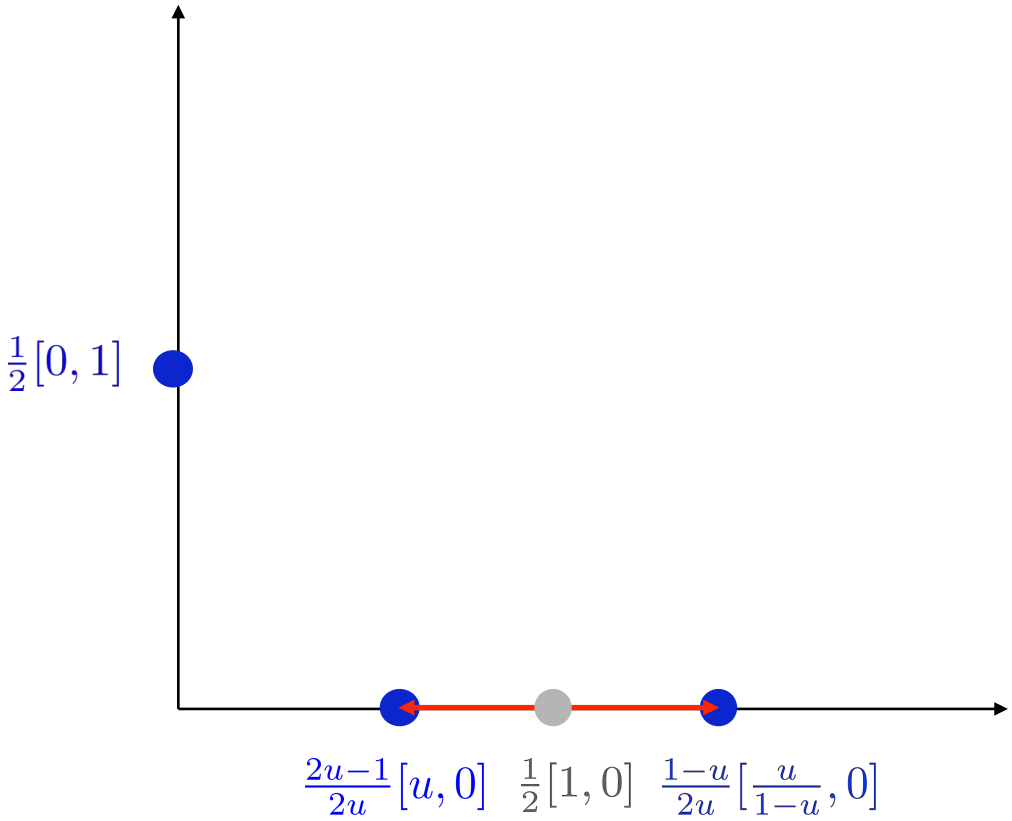
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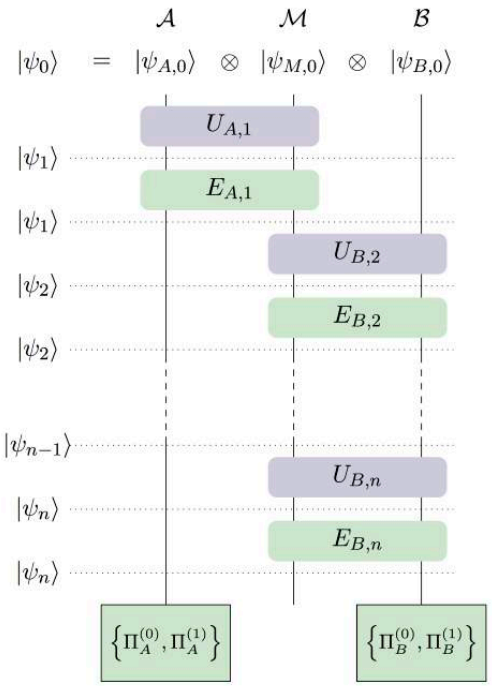


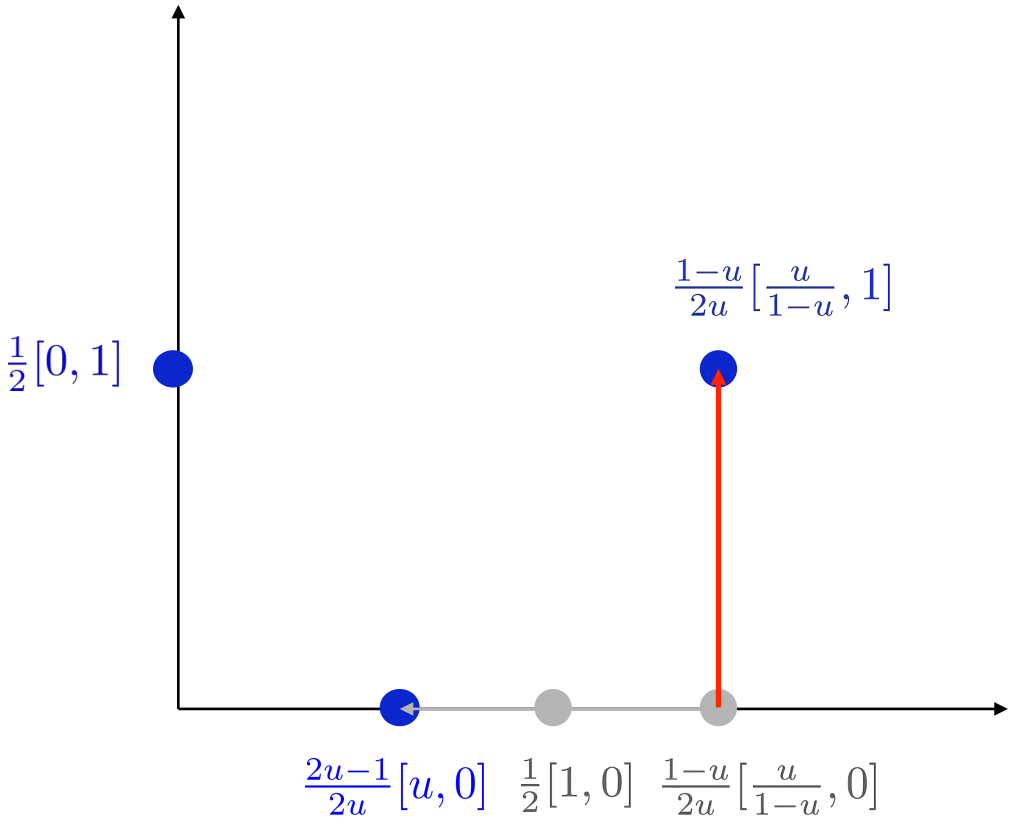
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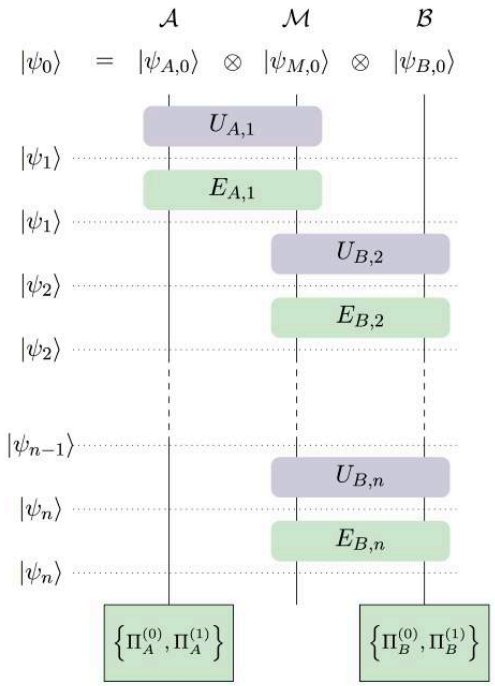


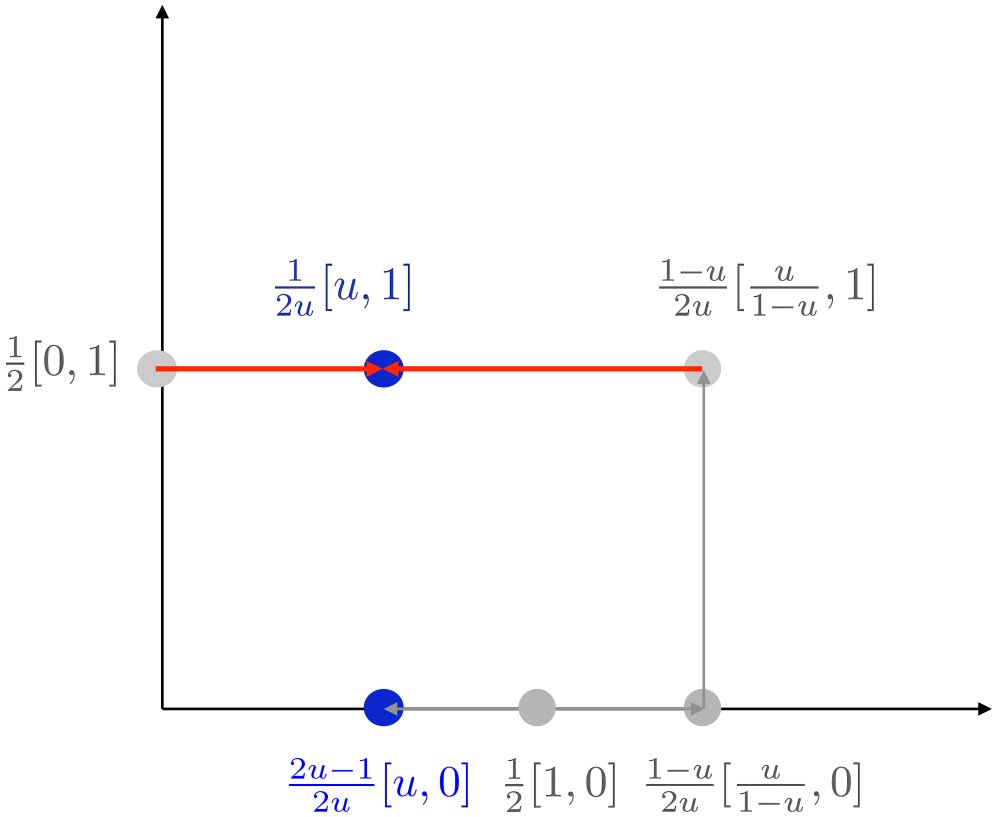
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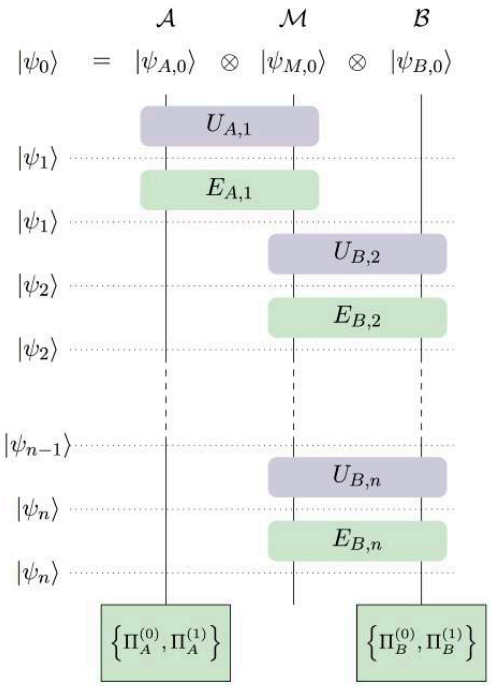


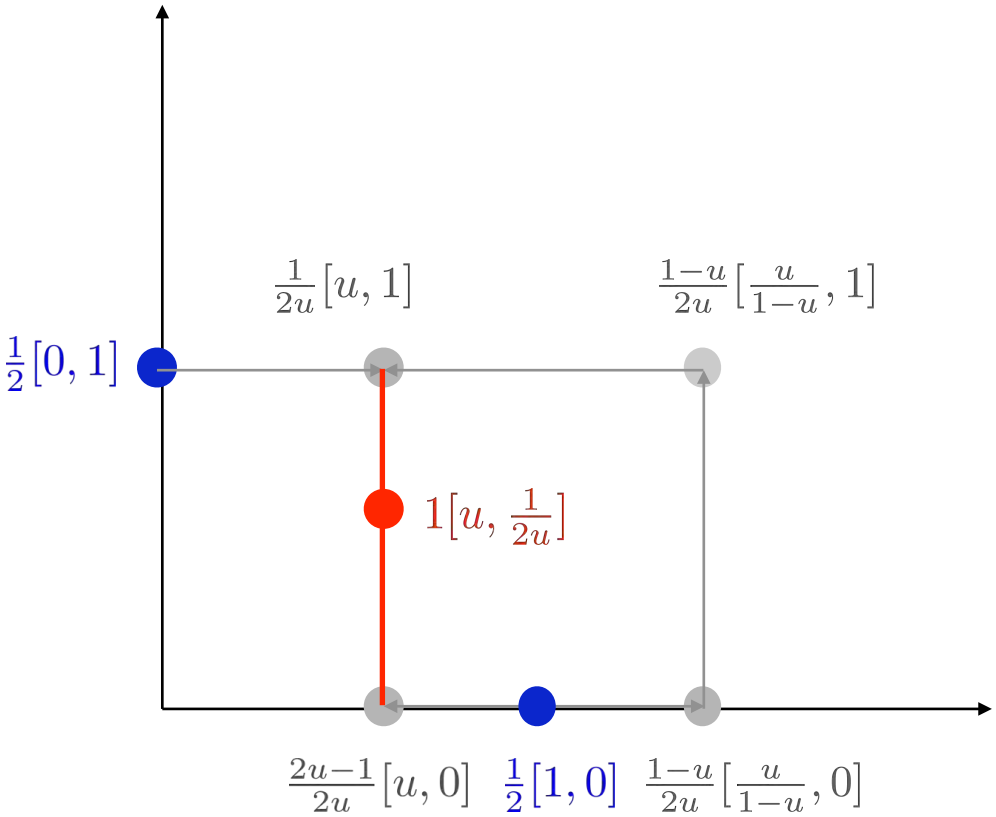
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3. Weak Coin Flipping: Point Games



Points and weights: from the dual feasible point and the honest states

$$\{Z_{A,i}\},\{Z_{B,i}\} \quad P_B^* \le \langle\psi_0| Z_{A,0} \otimes I_M \otimes I_B |\psi_0\rangle = \beta \qquad P_A^* \le \langle\psi_0| I_A \otimes I_M \otimes Z_{B,0} |\psi_0\rangle = \alpha$$

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Transitions: from the constraints of the dual SDP

3. Weak Coin Flipping: Protocols & Point Games

1. Equivalence of protocols and point games
   - Protocols and cheating strategies
   - Semi-definite programs and SDP duality
   - Topology of infinite dimensional convex cones
   - operator monotone functions
   - duality of convex cones

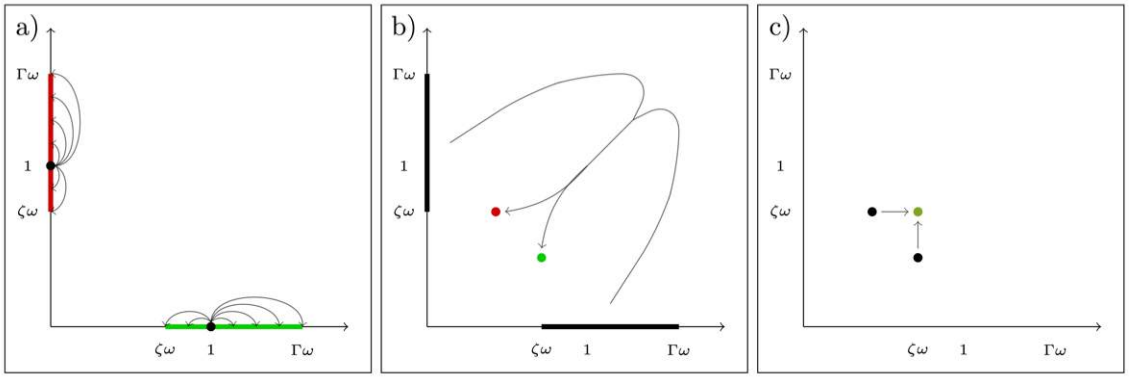
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2. Point game with final point [1/2+ε,1/2+ε] [Mochon]



3. Weak Coin Flipping: optimal protocols for SCF and BC

Assume a perfect weak coin flipping protocol WCF

Protocol

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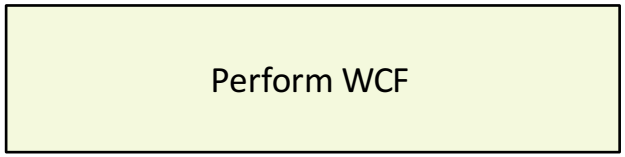
- Alice and Bob are honest: c is uniformly random.
- Alice is dishonest: If she wins the WCF then she chooses the value she wants, otherwise c is random

$$P_A^* = P_B^* = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$$

- Cheating player succeeds with prob. 1/2 even when he loses WCF

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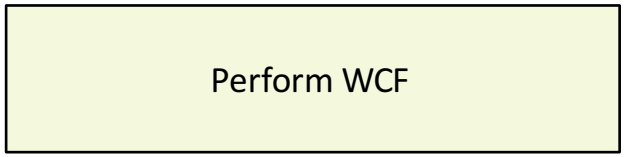
First Protocol



- Winner outputs a random c

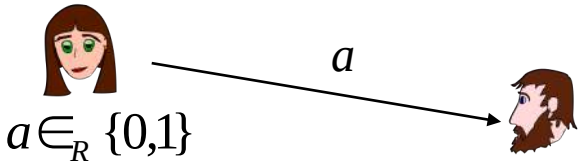
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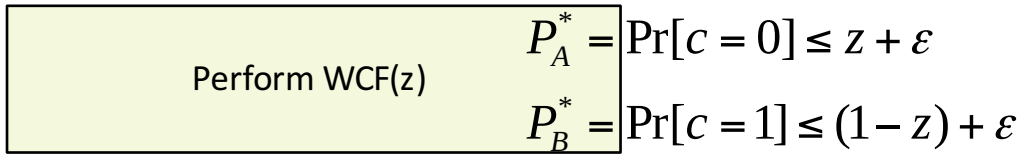
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New Protocol [Chailloux,K. 09]

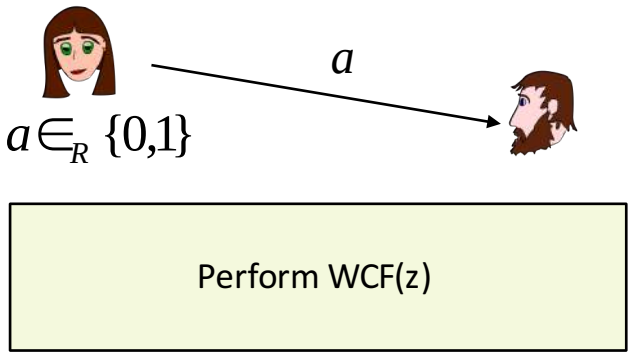




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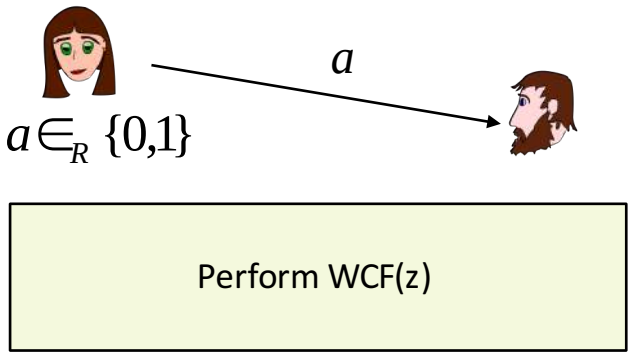
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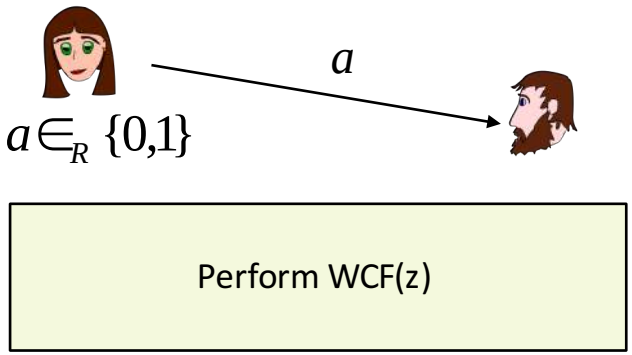
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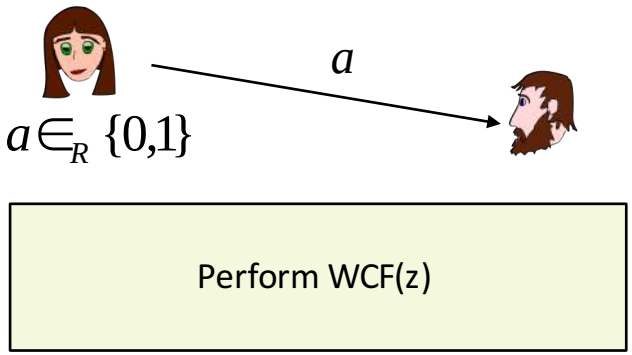
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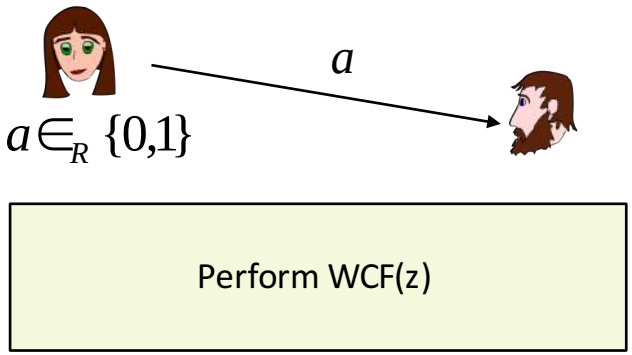
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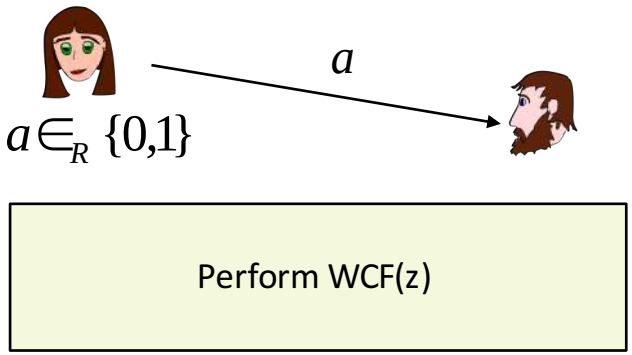
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- For $z = 2-\sqrt{2}$ we get the optimal protocol

1., 2. & 3. Recap

	Lower Bound	Upper Bound
Strong Coin Flipping	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$
Bit Commitment	0.739…	0.739…
Weak Coin Flipping	1/2	1/2+ε

– Optimal protocols: Trace distance/Fidelity, Semi-definite Programs, Point Games

– Weak security but strictly better than classical and forever

1., 2. & 3. Implementations

WCF: almost perfect security, but no idea what the protocol looks like

SCF and BC: very weak security, but implementable on top of other schemes

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Alice sends the second system to Bob.

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- Alice reveals x and sends the first system
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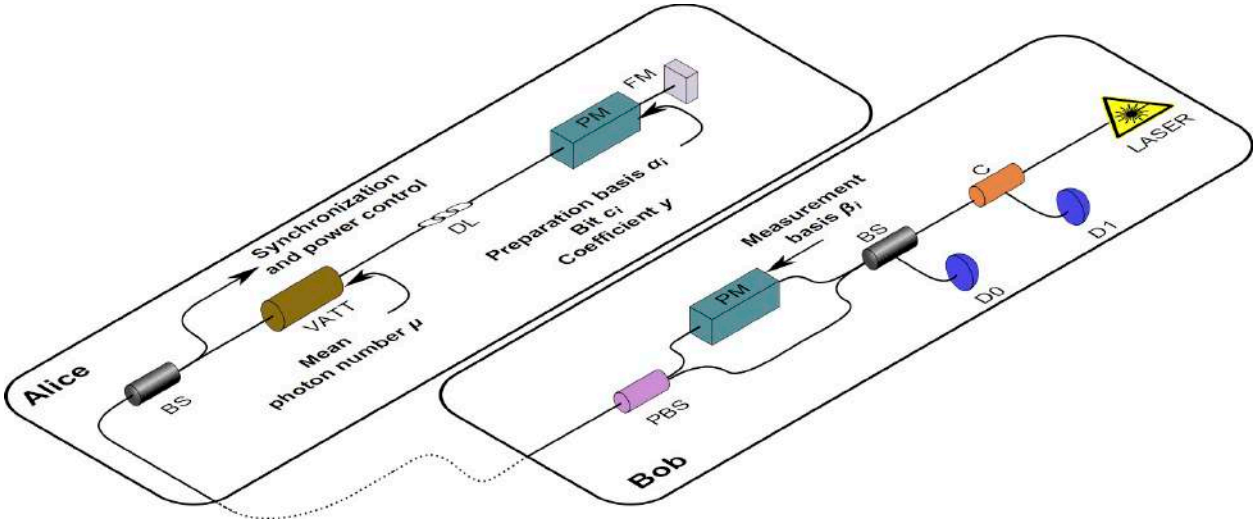
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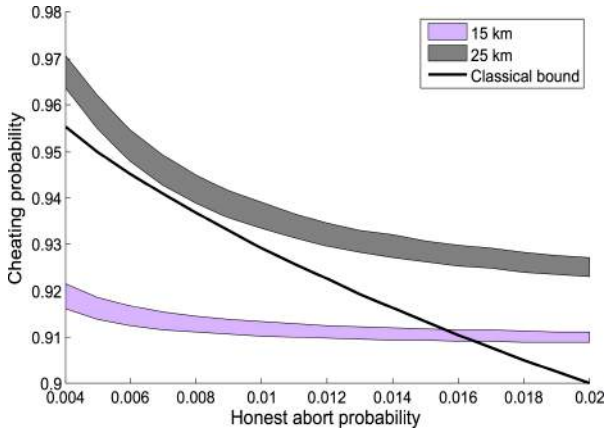
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Implementation over 15km with security provably higher than classical

[Pappa, Jouguet, Lawson, Chailloux, Legre, Trinkler, K., Diamanti, *Nature Communications* 14]





Outline

1. Bit Commitment (BC)
   a) A simple protocol with cheating probability ¾
   b) A lower bound via Fidelity/Trace Distance
2. Strong Coin Flipping (SCF)
   a) Bit-commitment based coin flipping with ¾
   b) A lower bound via Semi-Definite Programming
   c) Implementations
3. Weak Coin Flipping (WCF)
   a) Kitaev's point game formalism
   b) Optimal protocols for BC and SCF
4. More security models
   a) Device Independent: BC via the GHZ game
   b) Noisy storage: Weak String Erasure
   c) Relativistic: BC protocol
5. More primitives

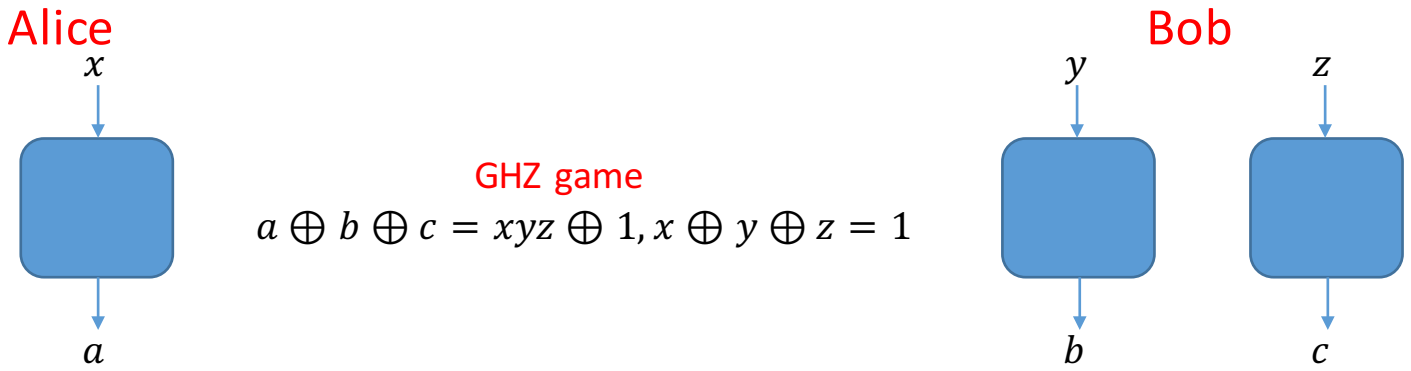
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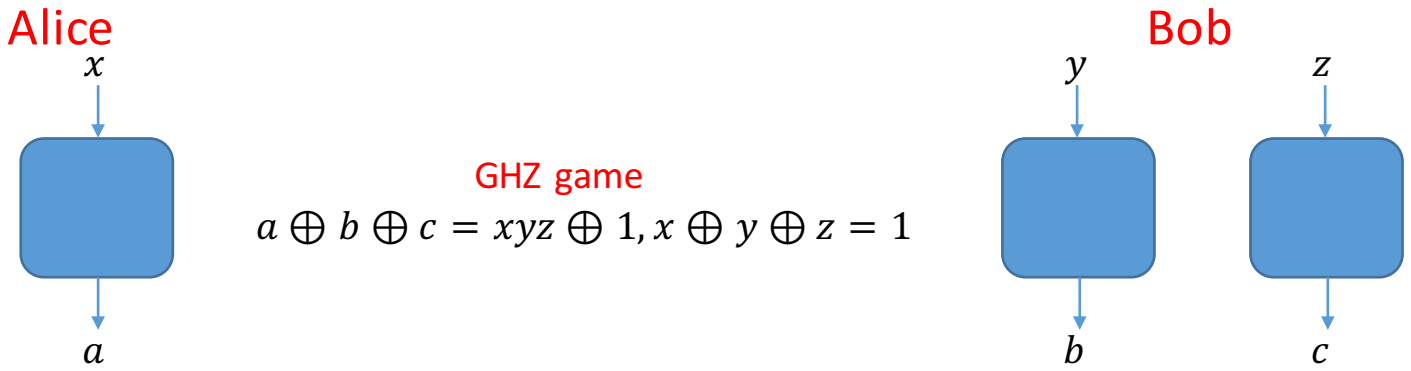
Device Independent Bit Commitment & Coin Flipping [Silman, Chailloux, Aharon, K., Pironio, Massar 11]



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Device Independent Bit Commitment & Coin Flipping [Silman, Chailloux, Aharon, K., Pironio, Massar 11]



Commit: Alice inserts x in her box and sends $c = a \oplus r \cdot x$, for a random r

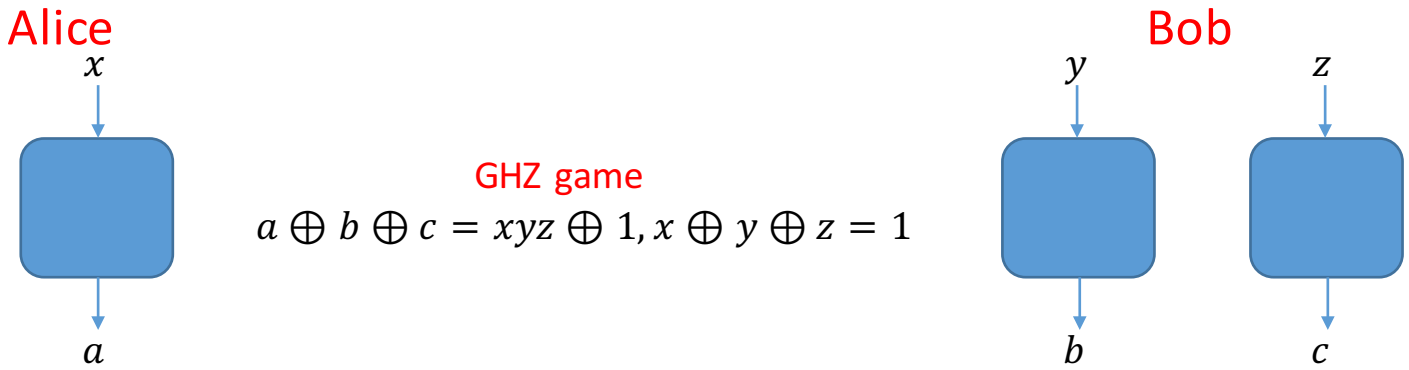
Reveal: Alice sends x, a.

Bob checks if c is consistent. Then picks $y, z: y \oplus z = x \oplus 1$and checks the GHZ condition

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Device Independent Bit Commitment & Coin Flipping [Silman, Chailloux, Aharon, K., Pironio, Massar 11]



Cheating Alice: $P_A^* = cos^2\theta$

Cheating Bob: $P_B^* = \frac{3}{4}$

Remark: the optimal cheating strategies do not tamper with the honest devices

4. Security models: Computational Assumptions

Classical Assumptions

- Factoring, Discrete Log
- Lattice Problems
- Hardware assumptions: One-time memories
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Quantum assumptions

- Lattice Problems?
- Can we base cryptography on QMA?
- One-time memories
- Noisy storage

4. Security models: Noisy Storage

Model: States in quantum memory deteriorate over time

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Main idea: Weak String Erasure

- Alice sends n BB84 states
- Bob measures in random bases.
- Wait for time t.
- Alice reveals bases.
- Output: Alice has an n-bit string. Bob has a subset of the string, which Alice does not know

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Implementations [Ng, Joshi, Ming, Kurtsiefer, Wehner 14]

Remarks

- Almost perfect security
- Noisy memory vs. factoring

4. Security models: Relativistic protocols

Main idea: Split Alice and Bob in two spatially separated agents that cannot communicate for some time due to relativity.

See talk on Friday

4. Security models: Relativistic protocols

Main idea: Split Alice and Bob in two spatially separated agents that cannot communicate for some time due to relativity.

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Implementations: [Lunghi, Kaniewski, Bussières, Houlmann, Tomamichel, Wehner, Zbinden 15]

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Remarks

- We can use classical messages
- Can we prove security against quantum adversaries?
- Can we justify the spatial split of the agents?

Outline

1. Bit Commitment (BC)
   a) A simple protocol with cheating probability ¾
   b) A lower bound via Fidelity/Trace Distance
2. Strong Coin Flipping (SCF)
   a) Bit-commitment based coin flipping with ¾
   b) A lower bound via Semi-Definite Programming
   c) Implementations
3. Weak Coin Flipping (WCF)
   a) Kitaev's point game formalism
   b) Optimal protocols for BC and SCF
4. More security models
   a) Device Independent: BC via the GHZ game
   b) Noisy storage: Weak String Erasure
   c) Relativistic: BC protocol
5. More primitives

5. More primitives

- Oblivious Transfer
  - Optimal bound in Information theoretic model is not known
  - A device independent protocol is not known
  - A noisy-storage protocol is known and implemented
  - A relativistic protocol is not known

- Positioned based crypto
- Zero Knowledge
- Money schemes
- Entanglement Verification
- Secret Sharing
- Delegation
- …

Open Problems

- Why do BC and CF have these security parameters?
- Understand/Implement Weak Coin Flipping
- Understand Oblivious Transfer (universal primitive)
- Will any of these protocols be deployed in practice?
- Multiparty primitives